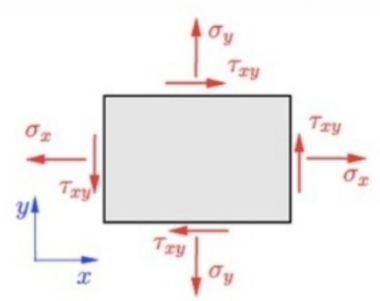


Homework 2.1 In a thin metal sheet the stresses σ_x , σ_y , τ_{xy} are given. Determine value and direction of the principal stresses.

Given: $\sigma_x = 20 \text{ MPa}$, $\sigma_y = 30 \text{ MPa}$, $\tau_{xy} = 10 \text{ MPa}$.



Direction of principal stresses:

$$\tan 2\varphi^* = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} = \frac{2(10) \text{ MPa}}{(20 - 30) \text{ MPa}} = \frac{20}{-10} = -2$$

$$\varphi_1^* = 58.28^\circ \quad \varphi_2^* = \varphi_1^* + \frac{\pi}{2} = 148.28^\circ$$

Value of principal stresses:

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

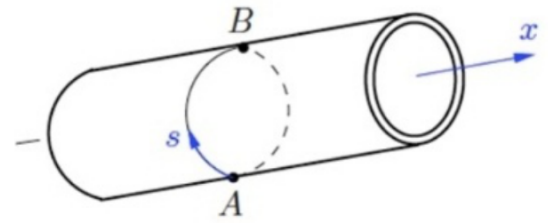
$$= 25 \pm \sqrt{25 + 100} = 25 \pm 5\sqrt{5}$$

$$\sigma_1 = 36.18 \text{ MPa} \quad \sigma_2 = 13.82 \text{ MPa}$$

Homework 2.2 A thin-walled tube is loaded by a bending moment, an internal pressure, and a torsional moment. In points A and B the following stresses occur due to the loading:

$$\sigma_{x_s}^{A,B} = \pm 50 \text{ MPa}, \sigma_s^{A,B} = 100 \text{ MPa}, \tau_{xs}^{A,B} = 100 \text{ MPa}.$$

Determine value and direction of the principal stresses in the points A and B.



For point A:

$$\begin{aligned} \text{Principal stresses } \sigma_{1,2} &= \frac{1}{2}(\sigma_x + \sigma_y) \pm \sqrt{\left[\frac{1}{2}(\sigma_x - \sigma_y)\right]^2 + \tau_{xy}^2} \\ &= 75 \pm \sqrt{25^2 + 100^2} \end{aligned}$$

$$\sigma_1 = 178.08 \text{ MPa}$$

$$\sigma_2 = -28.08 \text{ MPa}$$

Principal direction,

$$\begin{aligned} \tan 2\varphi^* &= \frac{2\tau_{xy}}{\sigma_x - \sigma_y} = \frac{200}{-50} = -4 \\ \varphi_1^* &= -52.02^\circ \quad \varphi_2^* = 142.02^\circ \end{aligned}$$

Point B:

$$\sigma_{1,2} = 25 + \sqrt{(-75)^2 + 100^2}$$

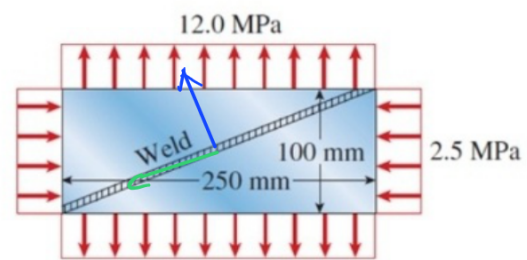
$$\sigma_1 = 150 \text{ MPa}$$

$$\sigma_2 = -100 \text{ MPa}$$

$$\tan 2\varphi^* = \frac{200}{-150} = -\frac{4}{3}$$

$$\varphi_1^* = 63.43^\circ \quad \varphi_2^* = 153.43^\circ$$

Homework 2.3 A rectangular plate is formed by welding two triangular plates (s. figure). Determine the normal stress σ_w acting perpendicular to the line of the weld and the shear stress τ_w acting parallel to the weld.



Transformation:

$$\sigma_{\xi} = \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2}(\sigma_x - \sigma_y)\cos 2\varphi + \tau_{xy}\sin 2\varphi$$

$$\sigma_{\eta} = \frac{1}{2}(\sigma_x + \sigma_y) - \frac{1}{2}(\sigma_x - \sigma_y)\cos 2\varphi - \tau_{xy}\sin 2\varphi$$

$$\tau_{\xi\eta} = -\frac{1}{2}(\sigma_x - \sigma_y)\sin 2\varphi + \tau_{xy}\cos 2\varphi$$

$$\sigma_x = -2.5 \text{ MPa} \quad \sigma_y = 12 \text{ MPa} \quad \tau_{xy} = 0$$

$$\sigma_{\xi} = \frac{1}{2}(-2.5 + 12) + \frac{1}{2}(-2.5 - 12)\cos(43.6) + 0$$

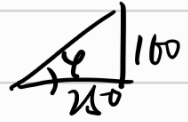
$$= -0.5 \text{ MPa}$$

$$\tau_{\xi\eta} = -\frac{1}{2}(-2.5 - 12)\sin(43.6) + 0$$

$$= 5 \text{ MPa}$$

$$\sigma_w + \sigma_{\xi} = \sigma_x + \sigma_y$$

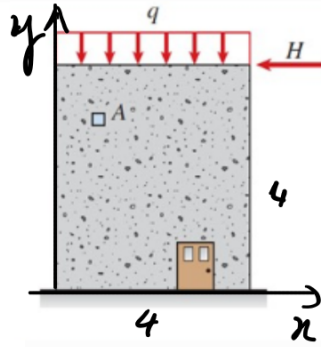
$$\sigma_w = \sigma_x + \sigma_y - \sigma_{\xi} = -2.5 + 12 + 0.5 = 10 \text{ MPa}$$



$$\tan \varphi = \frac{100}{250}$$

$$\varphi = 21.8^\circ$$

Homework 2.4 A shear wall in a reinforced concrete building (Width 4m, thickness 0.3m, height 4m) is subjected to a vertical uniform load of intensity $q=2\text{MPa}$ and a horizontal force $H=800\text{kN}$, as shown in the figure. Determine (a) the principal stresses; (b) the maximum shear stresses and associated normal stresses.



$$a) \sigma_x = 0 \quad \sigma_y = -2\text{MPa} \quad \tau_{xy} = \frac{H}{A} = \frac{800 \times 10^3}{1.2} = 0.6667\text{MPa}$$

$$\text{Principal stresses } \sigma_{1,2} : \frac{1}{2}(\sigma_x + \sigma_y) \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

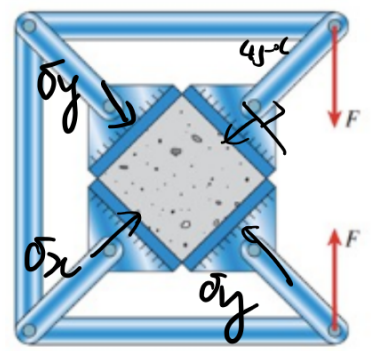
$$= -1 \pm \sqrt{1 + \left(\frac{2}{3}\right)^2}$$

$$\sigma_1 = 0.202\text{MPa} \quad \sigma_2 = -2.202\text{MPa}$$

$$b) \tau_{\max} = \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = 1.202\text{MPa}$$

$$\sigma_{\tau_{\max}} = \frac{\sigma_x + \sigma_y}{2} = -1\text{MPa}$$

Homework 2.5 A 100mm cube of concrete is compressed in biaxial stress by means of a framework that is loaded as shown in the figure. Assuming that each load F equals 200 kN, determine the maximum normal stresses and maximum shear stress in the concrete.



$$F \cos 45^\circ = 200 \times 10^3 \cos 45^\circ = 141.4214 \text{ kN}$$

$$\sigma_x = - \frac{141.4214 \times 10^3}{0.01} = -14.1421 \text{ MPa} = \sigma_y$$

(negative as they are compressive stresses)

The resolution of force F .

$$\tau_{xy} = \frac{F \sin 45^\circ}{A} = 14.1421 \text{ MPa}$$

We should have no shear stress on xy direction, since the friction on the contact surface is neglected.

$$\sigma_{1,2} = \frac{1}{2}(\sigma_x + \sigma_y) \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

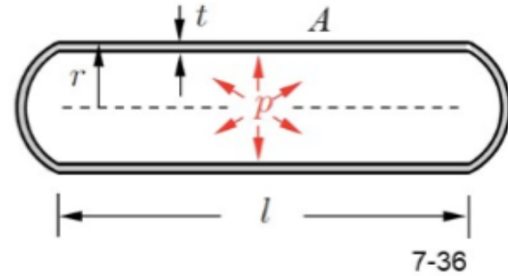
$$= \frac{1}{2}(-28.2842) \pm \sqrt{14.1421^2}$$

$$\sigma_1 = -4 \times 10^{-5} \text{ MPa} \quad \sigma_2 = -28.2842 \text{ MPa}$$

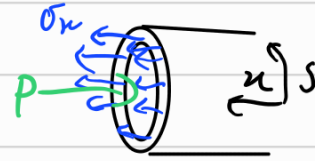
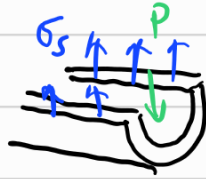
$$= -40 \text{ Pa}$$

$$\tau_{max} = \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \pm 14.1421 \text{ MPa}$$

Homework 2.6 A thin-walled cylindrical pressure vessel made of steel is subjected to an internal pressure p . What is the maximum value of the pressure such that the maximum shear stress in the central parallel pipe part do not to exceed the limit stress τ_{lim} ?



From Equilibrium conditions,



$$\sigma_x \cdot A_1 - p \cdot A_2 = 0$$

$$A_1 = \pi(t+r)^2 - \pi r^2 \quad \because t \ll r$$

$$\sigma_x \cdot 2\pi r t - p \pi r^2 = 0$$

$$\cancel{\pi t^2} + 2\pi t r + \cancel{\pi r^2} - \pi r^2 = 2\pi t r$$

neglected

$$\sigma_x = \frac{p r}{2t}$$

$$\sigma_\phi \cdot A_2 - p \cdot A_1 = 0$$

$$\sigma_\phi \cdot 2\pi r t - p \cdot 2\pi r R = 0$$

$$\sigma_\phi = \frac{p r}{t}$$

σ_x and σ_ϕ are principal stresses as shear stresses vanish in these sections.

$$\tau_{\text{lim}} > \tau_{\text{max}} = \pm \frac{1}{2}(\sigma_1 - \sigma_2) = \frac{1}{2} \left(\frac{p r}{t} \right)$$

$$p < \frac{4 t \tau_{\text{lim}}}{r}$$